J-C-145b
July 26, 1963
SUPERSEDING
Int. Fed. Spec. J-C-00145a (GSA-FSS)
October 9, 1962 and
Fed. Spec. J-C-145
October 26, 1945

FEDERAL SPECIFICATION

CABLE, POWER, ELECTRICAL AND WIRE, ELECTRICAL, (WEATHER-RESISTANT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers copper or aluminum, single conductor, solid or stranded, weather-resistant cable and wire for use in insulator-supported exterior applications.

1.2 Classification.

1.2.1 *Types, classes, and kinds.* Cable and wire furnished under this specification shall be of the following types, classes, and kinds, as specified (see 6.1).

Types:

Type I.—Neoprene covered.
Type II.—Polyethylene covered.
Type III.—Thermosetting polyethylene covered.

Classes:

Class 1.—Solid conductor (AWG 12—AWG 2 only).
Class 2.—Concentric-lay-stranded conductors of standard flexibility, conforming to class A of ASA C7.8 or C7.21

Kinds:

Kind A.—Aluminum conductor.
Kind C.—Copper conductor.

1.2.2 *Sizes.* Cable and wire furnished under this specification shall be of the following sizes, as specified (see 6.1).

Kind A.—AWG 8 through 500 MCM.
Kind C.—AWG 12 through 500 MCM.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

J-C-98—Cable and Wire, Insulated; Method of Sampling and Testing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a sub-

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scription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-C-12000—Cable, Cord And Wire, Electric, Packaging of.

Military Standard:

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Standards Association, Inc., Standards:

C7.8—Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft, Specification for.

C7.21—Concentric-Lay-Stranded Aluminum Conductors, Hard, Three-Quarter Hard, and Half Hard, Specification for.

C8.34—Weather-Resistant Wire and Cable, Neoprene-Type, Specification for.

C8.35—Weather-resistant Wire and Cable, Polyethylene-Type, Specification for.

(Copies may be obtained from the American Standards Association, Inc., 10 East 40th Street, New York 16, N. Y.)

3.1 Material. Cable and wire shall be as specified herein.

3.2 Type I cable and wire. Unless otherwise specified (see 6.1), type I cable and wire shall conform to the requirements of ASA C8.34.

3.3 Type II cable and wire. Unless otherwise specified (see 6.1), type II cable and wire shall conform to the requirements of ASA C8.35.

3.4 Type III cable and wire.

3.4.1 *Construction.* Type III cable and wire shall consist of copper or aluminum conductors conforming to the requirements of ASA C7.8 or C7.21, as applicable, and covered with a thermosetting polyethylene compound capable of continuous operation at conductor temperatures of 90°C. (194°F.) without degradation.

3.4.2 *Covering.*

3.4.2.1 *Thickness of covering.* The average thickness of the covering, considered as one-half the difference between the average overall diameter measured over the covering and the diameter of the bare conductor, shall be not less than that shown in table I. The minimum thickness at any point shall be not more than 5 mils below the specified average for thicknesses 3/64 inch and under, and not more than 10 percent below the specified average for thicknesses greater than 3/64 inch.

TABLE I.—*Covering thickness*

Size	Thickness
	64ths Inch
AWG 12 thru AWG 10	2
AWG 8 thru AWG 2	3
AWG 1 thru AWG 4/0	4
250 MCM thru 500 MCM	5

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3.4.2.2 *Physical properties.* The physical properties of the covering shall conform to the requirements of table II, when tested as set forth in 4.4.2.2.1 through 4.4.2.2.3.

TABLE II.—*Physical properties*

Initial:	
Tensile strength	1500 p.s.i.
Elongation	250 percent
Accelerated aging:	
Tensile strength	750 p.s.i.*
	900 p.s.i.**
Elongation	125 percent*
	150 percent**

* After air pressure-heat test.

** After air oven test and oil immersion test.

3.4.2.3 *Moisture absorption.* The moisture absorption of the covering shall not exceed 5 milligrams per square inch when tested as set forth in 4.4.2.3.

3.4.2.4 *Horizontal flammability.* The covering shall not burn for a distance greater than 2 inches to either side of the center point of application of the test flame when tested as set forth in 4.4.2.4. In addition, the duration of flaming of the sample after removal of the test flame shall not exceed 1 minute.

3.4.3 *Workmanship.* All material shall be new, suitable for the purpose intended, and free of any defects which impair the serviceability of the product.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling.**

4.2.1 *Lot.* Unless otherwise specified (see 6.1), the lot shall consist of the total number of coils or reels of the same type, class, kind, and size which are within a quantity offered for inspection at any one time or which are within a single shipment.

4.2.2 *Sampling for inspection and tests.* Unless otherwise specified (see 6.1), the following number of sample coils or reels shall be taken at random from each lot.

Total in the lot		Number of samples
Coils	*Reels*	
1	1	1
2	2	2
3 to 25	3 to 30	3
For each 25 additional	For each 10 additional	1 additional

4.2.2.1 A test sample shall be cut from each sample coil or reel. Each test sample shall be six feet long and shall be cut from the end of the sample coil or reel, disregarding the first two feet. For weight checks, an additional 2-foot sample is required.

4.3 **Visual inspection.** The visual inspection of each sample coil or reel shall cover construction, compactness, stranding, splices, joints, and repairs.

4.4 **Tests.**

4.4.1 *Types I and II cable or wire.* Each sample coil or reel shall be subjected to the tests set forth in ASA Standard C8.34 or C8.35, as applicable, to determine conformance to the requirements of 3.2 and 3.3, respectively.

4.4.2 *Type III cable or wire.* Type III cable or wire shall be subjected to the following tests, in accordance with the specified test method of J-C-98, to determine compliance with 3.4.

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4.4.2.1 *Thickness of covering.*

4.4.2.1.1 *Less than 3/64 inch.* Covering thickness shall be determined as set forth in method 1014 of J-C-98.

4.4.2.1.2 *Nominal 3/64 inch and over.* Covering thickness shall be determined as set forth in method 1011 of J-C-98.

4.4.2.2 *Physical properties of covering.*

4.4.2.2.1 *Tensile strength.* Tensile strength, initially and after accelerated aging, shall be determined as set forth in method 3021 of J-C-98.

4.4.2.2.2 *Elongation.* Elongation, initially and after accelerated aging, shall be determined as set forth in method 3031 of J-C-98.

4.4.2.2.3 *Accelerated aging.*

4.4.2.2.3.1 *Air pressure-heat test. The air* pressure-heat test shall be conducted in accordance with method 4021 of J-C-98, except that the aging period shall be 40 plus or minus 1/4 hours rather than 20 plus or minus 1/4 hours.

4.4.2.2.3.2 *Air oven test.* The air oven test shall be conducted in accordance with method 4031 of J-C-98, except that the test temperature shall be 121° plus or minus 1°C. rather than 70° plus or minus 1°C.

4.4.2.2.3.3 *Oil resistance test.* The oil resistance test shall be conducted in accordance with method 4221 of J-C-98.

4.4.2.3 *Moisture absorption.* The moisture absorption characteristic of the covering shall be determined in accordance with method 8211 of J-C-98.

4.4.2.4 *Flammability.* The horizontal flammability of the covering shall be determined in accordance with method 5211 of J-C-98.

4.5 Rejection. If 20 percent or more of the samples taken from a lot fail to meet the requirements of this specification, the lot shall be rejected. However, at the contractor's option, a second group of samples may be taken at random from the lot and tested at the contractor's expense. If every sample of the second group meets the requirements of this specification, the lot shall be considered acceptable.

4.5.1 In any event, individual coils or reels which are found to fail to meet the requirements of this specification shall be rejected.

4.6 Preparation for delivery inspection. The packaging, packing, and related marking of all cable and wire produced for delivery under the contract shall be inspected to verify conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Packaging and packing. Packaging and packing of electrical cable, cord and wire shall be in accordance with MIL-C-12000, as specified for level A, B, or C (see 6.1). Level A packaging requirements shall be used for civil agency procurements when level B shipments and storage are required.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein,

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and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.

(*b*) Type, class, kind, and size of cable and wire (1.2).

(*c*) Cable and wire requirements, if other than specified (3.2 and 3.3).

(*d*) Lot formation, if other than specified (4.2.1).

(*e*) Sampling requirements, if other than specified (4.2.2).

(*f*) Level of packaging and packing required (5.1).

(*g*) Special marking, if required (5.2).

6.2 If laboratory tests by the Government are required, purchasing officers should order cable and wire for test purposes in addition to the number of feet required for installation.

6.3 **Transportation description.** Transportation description and minimum weights applicable to this commodity are:

Rail—Aluminum:

Cable, Electric, Aluminum.
Carload minimum weight 30,000 pounds.

Motor:

Cable, Electric Aluminum.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Rail—Aluminum:

Wire, Aluminum, Plain or Insulated.
Carload minimum weight 30,000 pounds.

Motor:

Wire, Aluminum, Plain or Insulated.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Rail—Copper:

Cable, Electric, Copper NOIBN.
Carload minimum weight 30,000 pounds.

Motor:

Cable, Electric, Copper, NOI
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Rail—Copper:

Wire, Copper, Insulated or Plain.
Carload minimum weight 30,000 pounds.

Motor:

Wire, Copper, Insulated or Plain.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

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MILITARY INTEREST:
Army—EL
Navy—Y
Air Force—RADC

Copies of this specification may be purchased for cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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